HyperSF: A Hypergraph Representation Learning Method Based on Structural Fusion

Xiangfei Fang[1, 2], Chengying Huan[3*], Boying Wang[4], Shaonan Ma[5], Heng Zhang[1], Chen Zhao[1]
[1]Institute of Software, Chinese Academy of Sciences,[2]University of Chinese Academy of Sciences,
[3]Nanjing University, [4]North University of China, [5]Qiyuan Lab

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***Abstract*—Hypergraph Neural Networks (HNNs) have recently gained attention as a powerful approach for capturing high-order correlations through hypergraph-structured encoding and learning techniques. However, despite their potential, existing HNN methods often encounter over-smoothing issues, which limit their ability to effectively integrate global information while maintaining high-order structural details. This limitation compromises the overall effectiveness of these models. To tackle this challenge, we introduce a novel HNN framework called Hypergraph Structural Fusion (*HyperSF*). *HyperSF* combines the structural characteristics of both hypergraphs and graphs to effectively integrate global and local information while preserving the complex high-order structures inherent in hypergraphs. This structural fusion mechanism significantly improves model performance by ensuring that both types of information are utilized in a balanced manner. Comprehensive evaluations show that our method outperforms state-of-the-art approaches, demonstrating its effectiveness in hypergraph representation learning.**

***Index Terms*—Hypergraph Representation Learning, Structural Fusion.**

I. Introduction

Hypergraphs, which extend traditional graphs by allowing edges (hyperedges) to connect multiple nodes, provide a powerful mathematical framework for representing complex, high-order relationships involving multiple entities. This approach has proven effective in diverse fields such as social networks [1]–[3], computer vision [4]–[6], and natural language processing [7]–[9]. Building on the success of Graph Neural Networks (GNNs), there has been increasing interest in adapting GNN methodologies to hypergraphs, leading to the development of Hypergraph Neural Networks (HNNs).

Recent HNNs adopt direct message-passing mechanisms to learn node feature representations [10]–[12]. This process is typically divided into two stages: first, information is propagated from nodes to hyperedges; then, it is relayed back from hyperedges to nodes. In this framework, the hypergraph structure aggregates node information at their associated hyperedges, which is subsequently combined and distributed back to all nodes within each hyperedge, thereby updating their feature representations.

However, this message-passing mechanism has notable drawbacks, particularly the problem of **oversmoothing**. As each hyperedge can contain multiple nodes, the number of higher-order neighbors in a hypergraph grows rapidly, causing the global information to become less distinguishable as the model depth increases. Consequently, existing HNNs generally use only one or two layers, limiting nodes to information from their immediate hyperedge-connected neighbors.

To address these challenges, we propose an innovative framework called Hypergraph Structural Fusion (*HyperSF*). This framework dynamically aggregates global and local information without compromising the high-order structural integrity of the original hypergraph. Moreover, our approach eliminates the need to stack multiple HNN layers to capture global information, thereby improving the speed of model training and inference. Our approach starts by extracting global information from the initial hypergraph. Specifically, we use information from higher-order neighbors and their corresponding induced subgraphs, known as k-girded subgraphs, to form a global structural representation, which is then transformed into a graph structure. Next, we introduce a novel structural fusion-based information aggregation module. In this module, a node retrieves global information from the distilled graph while simultaneously acquiring local information from the original hypergraph. Compared to existing methods, this aggregation module integrates global relationships to enhance the information aggregation process and effectively addresses the issue of information loss. Furthermore, we incorporate a graph structure learning strategy that utilizes adaptive global information updates to eliminate redundant global information. To evaluate the efficiency and effectiveness of *HyperSF*, we perform node classification tasks on eight real-world datasets and compare the results with state-of-the-art HNNs. Experimental results show that *HyperSF* consistently outperforms all benchmark methods across these datasets, demonstrating significant improvements in accuracy. Notably, *HyperSF* achieves a substantial increase in prediction accuracy across all datasets.

II. Background and Related Work

Hypergraph A hypergraph $HG = (\mathcal{V}, \mathcal{E})$ is defined by a set of nodess $\mathcal{V}$ and a set of hyperedges $\mathcal{E}$. The total number of nodes is denoted by $n = |\mathcal{V}|$. The set of hyperedges, $\mathcal{E}$, contains $m = |\mathcal{E}|$ hyperedges, where each hyperedge $e \in \mathcal{E}$ is a subset of $\mathcal{V}$, meaning $e \subseteq \mathcal{V}$.

Hypergraph Neural Networks. Recent research has explored efficient learning techniques for hypergraphs. Feng et al. [13] pioneered the use of convolution operators in

*Corresponding Author.

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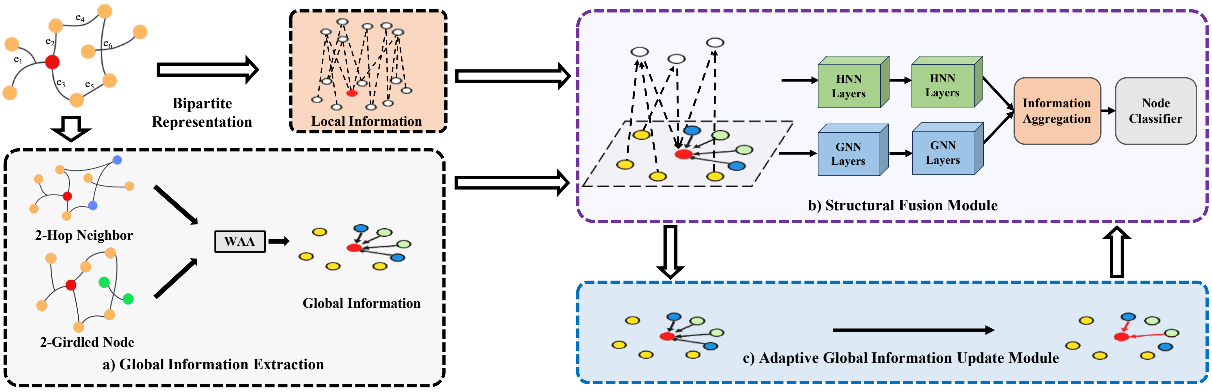

Fig. 1: Visual illustration of our proposed method, *HyperSF*, where WAA denotes the weighted arithmetic averaging operator. The method comprises three key components: a) the Global Information Extraction module, which extracts k-hop neighbors and k-girdled nodes from the original hypergraph structure and integrates them as global information; b) the Structural Fusion module for effective information aggregation; and c) the Adaptive Global Information Update module, which is responsible for removing redundant global information.

hypergraph learning by employing Laplacian matrices for HNN training. Several subsequent HNNs have been developed to convert hypergraphs into graphs and apply message-passing techniques to perform convolution operations on the resulting graph structures [14]–[17].

Diverging from these approaches, HyperSAGE [10] extends message-passing neural networks (MPNNs) to hypergraphs using a two-stage message-passing mechanism. Building on this, UniGNN [11] introduces the use of GCN [18], GAT [19], and GIN [20] models within hypergraph contexts. Further expanding on the two-stage message-passing paradigm, AllSet [12] incorporates a comprehensive set of message-passing models that handle information exchange between nodes and hyperedges. Building on AllSet's framework, methods such as HyperND [21], HNHN [22], HCHA [23], EDHNN [24], SheafHNN [25], and HyperGT [26] are essentially all two-stage message-passing networks. These methods further optimize HNN architectures and enhance model performance by incorporating mechanisms like attention within the two-stage message-passing framework.

III. Our Method: *HyperSF*

We propose our new model, named ***HyperSF***, to overcome the over-smoothing challenge, as illustrated in Fig. 1. *HyperSF* begins by extracting global structural information from the original hypergraph, which is represented as a graph structure (Section III-A). This global information is then combined with the original local information through a structural fusion process (Section III-B), effectively mitigating the **oversmoothing challenge**. During training, graph structure learning techniques are applied to eliminate redundant global information (Section III-C), retaining only the most relevant global details.

A. Global Information Extraction

In real-world applications of hypergraphs, the original inputs often lack explicit connections between nodes. Conventional methods, such as clique expansion [15], aim to derive graph connections and features from hypergraphs while minimizing information loss. In this work, we propose extracting crucial global information from multi-hop neighbors and their corresponding induced subgraphs. To establish the foundation for this approach, we introduce several definitions.

Definition 1. ***K-order Neighbors.*** *For a node v in the hypergraph H, the k-order neighbors, denoted $N^K_{v,H}$, are defined as the set of nodes in H that are reachable via paths from node v and the shortest path comprises exactly k hyperedges.*

Based on Definition 1, we define the notion of a k-girdled subgraph in H as follows:

Definition 2. ***K-girdled Subgraph.*** *A k-girdled subgraph, denoted as $H^K_{v,H} = (V^K_{v,H}, E(N^K_{v,H}))$, is the subgraph formed from hyperedges $E(N^K_{v,H})$ within the entire hypergraph H. These hyperedges include at least two k-order neighbors of v.*

The concept of a k-girdled node is defined as follows:

Definition 3. ***K-girdled Nodes.*** *The k-girdled nodes, represented as $Q^K_{v,H}$, are those nodes that are part of the k-girdled subgraph but are not k-order neighbors of v.*

Fig. 2 illustrates an example of a 2-girdled node and its corresponding subgraph. The blue nodes are identified as 2-girdled nodes, while the purple nodes represent 2-hop neighbors. The subgraph, depicted by the light green elliptical area, represents a 2-girdled subgraph.

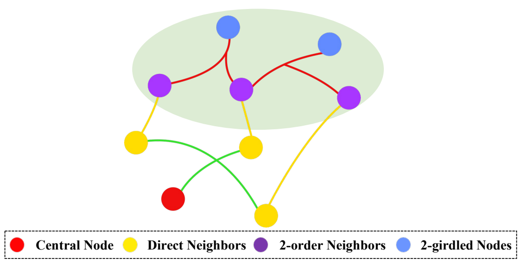

Fig. 2: An example of 2-girdled nodes and a 2-girdled subgraph of the red node.

Following the previously stated definition, global information is characterized as the weighted sum of the k-order neighbors and k-girdled nodes. Consequently, the aggregated global features of node v at layer l are defined as:

$$p_{G,v}^{l} = \sum_{k=2}^{m} \left(\sum_{u \in N_{v,H}^{k}} f_{k,N}^{l}(p_u^l) + \sum_{w \in Q_{v,H}^{k}} f_{k,Q}^{l}(p_w^l) \right), \quad (1)$$

where m denotes the maximum number of hops, and both $f_{k,N}^{l}$ and $f_{k,Q}^{l}$ function as aggregation operators. The adjacency matrix is given by $\mathbf{M} = \mathbf{H}\mathbf{H}^{\top}$. The global information can be expressed in matrix form as:

$$\mathbf{A} = \sum_{k=2}^{m} \left(\theta_{k,N}\mathbf{M}^{k}\right) + \sum_{k=2}^{m} \left(\theta_{k,Q}\mathbf{M}_{Q}^{k}\right), \quad (2)$$

where $\mathbf{M}_{Q}^{k} \in \mathbb{R}^{n \times n}$ is the girdle matrix such that the element at the m^{th} row and n^{th} column is non-zero, indicating that the n^{th} node is the k-girdled node of the m^{th} node. The weighting coefficients $\theta_{k,N}$ and $\theta_{k,Q}$ satisfy the condition $\sum_{k=2}^{m}(\theta_{k,N} + \theta_{k,Q}) = 1$, where both $\theta_{k,N}$ and $\theta_{k,Q}$ are within the range $[0, 1]$, to ensure the convergence of the matrix summation. The matrix $\mathbf{A}$ can be regarded as an adjacency matrix that incorporates information from higher-order neighbors.

B. Structural Fusion

In this subsection, we aggregate the extracted global information with the original local information by fusing graph and hypergraph structures, as illustrated in Fig. 1. We achieve Structural Fusion through two neural networks: a GNN network and an HNN network. Through structural fusion, a node in *HyperSF* can gather local information from direct neighbors via HNNs, while also obtaining global information from important remote neighbors through GNNs. Consequently, our *HyperSF* effectively overcomes the problem of oversmoothing, allowing it to capture meaningful global information.

For the hypergraph neural network, we employ D_v^{-1} and D_e^{-1} as normalization coefficients, just like HGNN^{+} [27], where D_v and D_e are respectively the diagonal matrices of nodes and hyperedges. The local information is then represented in matrix form as follows:

$$\mathbf{P}_{H}^{l+1} = \mathbf{D}_{v}^{-1}\mathbf{H}\mathbf{D}_{e}^{-1}\mathbf{H}^{\top}\mathbf{P}_{H}^{l}\Theta^{l}, \quad (3)$$

where Θ^{l} s a learnable parameter matrix at the l^{th} layer.

Meanwhile, nodes in *HyperSF* can also obtain global information from the auxiliary graph extracted in Subsection III-A. We define the adjacency matrix of the auxiliary graph as $\mathbf{A}$. The global information is represented using $\mathbf{A}$ in the following matrix form:

$$\mathbf{P}_{G}^{l+1} = \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}\mathbf{P}^{l}\mathbf{W}^{l}, \quad (4)$$

where $\mathbf{W}^{l}$ is a trainable parameter matrix and $\mathbf{D}$ is the degree matrix of $\mathbf{A}$. The pattern matrix $\mathbf{P}^{l+1}$ at the $l{+}1^{th}$ layer is obtained by aggregating the local features derived from the HNN and the global features obtained from the GNN, while retaining a portion of the original features $\mathbf{P}^{l}$ as:

$$\mathbf{P}^{l+1} = \sigma\left(\Gamma(\mathbf{P}^{l}, \mathbf{P}_{H}^{l+1}, \mathbf{P}_{G}^{l+1})\right), \quad (5)$$

where Γ represents the aggregation function in the l^{th} layer.

C. Adaptive Global Information Update

The adjacency matrix $\mathbf{A}$ may contain a significant amount of redundant information, which can contribute to the oversmoothing problem in the model. To address this, we employ a graph structure learning method to isolate essential global information, enabling the model to effectively train node features while eliminating noisy edges within the auxiliary graph. This process allows the neural network to adjust the graph structure dynamically during training, based on the initial adjacency matrix $\mathbf{A}$, thereby enhancing the model's accuracy. Our goal is to train a parameterized matrix $\hat{\mathbf{Z}}$ such that, in each iteration of *HyperSF*, the updated adjacency matrix is given by $\hat{\mathbf{A}} = \mathbf{A} - \hat{\mathbf{Z}}$. To prevent $\hat{\mathbf{A}}$ from becoming overly dense, we apply the $L1$ norm to limit the number of connected edges. A regularization term is added to the final loss function to penalize the $L1$ norm value of $\hat{\mathbf{A}}$.

For the final node classification task, we use the cross-entropy loss function. The complete loss function is structured as follows:

$$\mathcal{L} = \sum_{\mathrm{n}} (-(y\log(\hat{y}) + (1-y)\log(1-\hat{y})) + \lambda\|\hat{\mathbf{A}}\|, \quad (6)$$

where λ represents the regularization parameters. y and $\hat{y}$ respectively denote the actual labels and predicted labels of the nodes. By employing this strategy, nodes can obtain global information only from truly important higher-order neighbors in the auxiliary graph. Moreover, this approach allows for the adaptive adjustment of weights based on the relevance of the information without compromising the integrity of the original hypergraph structure.

IV. Evaluation

A. Experimental settings

Datasets. The evaluation is conducted on eight datasets, grouped into three distinct categories. The first category includes hypergraph-based citation networks: CocitationCora, CocitationPubmed, CocitationCiteseer, and CoauthorshipCora [14]. The second category comprises adaptations of House and Senate datasets [28], as proposed in [29]. The third category consists of two visual object classification datasets: ModelNet40 [30] and NTU2012 [31].

Baselines. Our model's performance is evaluated through classification tasks on hypergraph nodes. The baseline models include HGNN [13], HyperGCN [14], UniGNN [11], HNHN [22], HCHA [23], AllSet [12], HGNN^{+} [27], and EDHNN [24]. All these baselines have open-source implementations available, ensuring the replicability of results.

Experiment Settings. We follow the standard practice of partitioning each dataset into training, validation, and testing sets, maintaining a ratio of $2:1:1$. For optimization, we use the AdaGrad algorithm [32]. To ensure robustness and reliability, each model is trained and validated ten times with distinct configurations. The evaluation metric used is prediction accuracy, and all experiments are conducted on a server equipped with NVIDIA RTX 3090 GPUs.

TABLE I: Performance comparision with the state-of-the-art HNNs on hypergraph datasets. **Bold font** are used to indicate the best result, while the best baselines are underlined.

Methods	Cora	Pubmed	Citeseer	Cora-CA	House	Senate	ModelNet40	NTU2012
HGNN	78.98% ± 0.71	87.51% ± 0.35	73.07% ± 1.35	82.67% ± 1.91	68.27% ± 2.96	59.72% ± 4.61	95.27% ± 0.37	86.74% ± 1.31
HyperGCN	78.21% ± 2.09	83.68% ± 9.63	71.52% ± 1.28	79.90% ± 2.18	47.96% ± 2.18	52.68% ± 7.72	75.07% ± 3.26	53.76% ± 4.63
HNHN	75.92% ± 2.01	86.48% ± 0.53	72.13% ± 1.43	76.98% ± 2.14	75.48% ± 2.31	53.52% ± 6.04	97.19% ± 0.35	88.65% ± 1.30
HCHA	79.14% ± 1.32	86.72% ± 0.32	72.48% ± 1.36	83.03% ± 1.06	67.59% ± 2.99	51.63% ± 4.16	94.60% ± 0.29	87.41% ± 1.64
UniGNNII	78.94% ± 1.04	88.57% ± 0.49	73.13% ± 1.30	82.67% ± 1.36	68.61% ± 3.17	54.23% ± 5.68	97.50% ± 0.31	88.77% ± 0.89
HGNN$^+$	79.01% ± 0.91	87.45% ± 0.40	73.02% ± 1.43	82.84% ± 1.02	65.85% ± 2.85	55.63% ± 6.88	94.35% ± 0.39	86.66% ± 1.33
AllSetTransformer	78.64% ± 1.47	88.71% ± 0.35	73.08% ± 0.74	83.35% ± 1.41	76.34% ± 1.89	53.81% ± 5.30	98.11% ± 0.26	88.67% ± 1.29
AllDeepSets	76.65% ± 1.53	88.72% ± 0.48	70.08% ± 1.72	81.08% ± 1.40	76.47% ± 1.89	56.90% ± 5.22	96.81% ± 0.36	88.39% ± 1.25
EDHNN	80.04% ± 0.90	88.42% ± 0.36	74.03% ± 1.27	84.09% ± 1.59	73.02% ± 2.73	65.49% ± 2.02	97.28% ± 0.27	87.95% ± 0.94
***HyperSF* (ours)**	**81.15% ± 1.30**	**88.94% ± 0.28**	**75.01% ± 1.22**	**85.30% ± 1.49**	**77.55% ± 2.65**	**77.04% ± 3.87**	**98.52% ± 0.23**	**90.36% ± 1.08**

B. Experiment Results

Comparison with SOTA methods. Table I presents a comparative analysis of our *HyperSF* model against the ten hypergraph baselines mentioned above. The results reveal that while the baseline models exhibit varying strengths and limitations across different datasets, *HyperSF* consistently demonstrates superior performance, surpassing all state-of-the-art models in terms of accuracy.

The Senate dataset is particularly noteworthy, where *HyperSF* outperforms the second-best model by a significant margin of at least 2.0% on average. We attribute this notable improvement in accuracy to our structural fusion mechanism, which effectively aggregates global information with local information. This mechanism allows our model to address the over-smoothing problem without losing high-order information by using a single-layer HNN, thereby enhancing overall effectiveness. Additionally, our model shows strong performance across all four citation networks. Two key factors contribute to this success. First, our structural fusion mechanism efficiently aggregates global information without the need for multiple layers of HNNs. Second, we employ a graph structure learning method to dynamically update global information and remove redundant data from the auxiliary graphs, further improving overall performance.

TABLE II: Result of ablation experiments.

Dataset	w/o both	w/o AGIU	with both
Citeseer	73.90 ± 1.06	74.38 ± 1.33	75.10 ± 1.33
NTU2012	88.89 ± 1.08	89.74 ± 1.28	90.22 ± 1.12

Ablation Study. We analyze the effectiveness of the *Structural Fusion (SF)* module and the *Adaptive Global Information Update (AGIU)* module on performance by excluding them from the full model on two selected datasets. Specifically, we create a variant, *HyperSF* (w/o AGIU), by removing the AGIU module. Additionally, we combine the removal of both modules to obtain *HyperSF* (w/o both). As shown in Table II, *HyperSF* (w/o AGIU) yields inferior performance compared to *HyperSF* (with both), highlighting the critical role of adaptive global information update in removing redundant global information, which is crucial for effective classification. *HyperSF* (w/o both) performs the worst in the experiment, demonstrating that our structural fusion-based method significantly enhances discriminability.

Hyperparameter Analysis. We analyze the impact of λ on the experimental results. As shown in Fig. 3(a), when the value of λ increases from 10^{-5} to 10^{-2}, the accuracy initially improves and then stabilizes within a certain range. However, if λ continues to decrease beyond this range, the accuracy begins to decline. These results suggest that controlling the sparsity of $\hat{\mathbf{A}}$ to an appropriate extent can effectively filter out redundant high-order structural information, thereby preventing the oversmoothing problem.

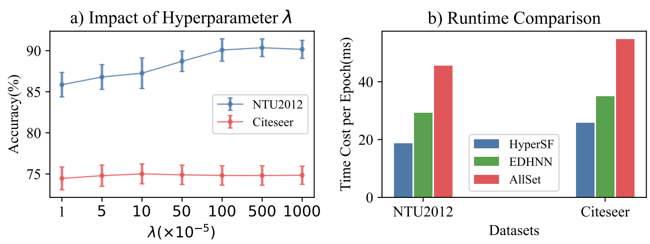

Fig. 3: Analysis of hyperparameter sensitivity and runtime.

Runtime Comparison. Finally, we compare the runtime of *HyperSF* with two top-performing methods, EDHNN and AllSetTransformer, as shown in Fig. 3(b). Our model outperforms these two models while also having lower computational costs. Notably, on the NTU2012 dataset, *HyperSF* achieves optimal performance with just one convolutional layer, significantly reducing the overall time overhead. This indicates that *HyperSF* can enhance accuracy while reducing time consumption to a certain extent.

V. Conclusion

In this work, we propose amplifying node distinction by integrating global information with local information. We introduce a novel hypergraph neural network framework, *HyperSF*, which leverages structural fusion to effectively integrate and enhance information from both direct and higher-order neighbors, thereby improving the performance of HNNs. Comprehensive experiments on real-world datasets underline the superiority of our proposed method compared to leading-edge counterparts. Overall, the results suggest that our model offers a notable improvement over existing methods.

References

[1] S. Wu, F. Sun, W. Zhang, X. Xie, and B. Cui, "Graph neural networks in recommender systems: a survey," *ACM Computing Surveys*, vol. 55, no. 5, pp. 1–37, 2022.

[2] T. R. Gwadabe and Y. Liu, "Improving graph neural network for session-based recommendation system via non-sequential interactions," *Neurocomputing*, vol. 468, pp. 111–122, 2022.

[3] T. Huang, Y. Dong, M. Ding, Z. Yang, W. Feng, X. Wang, and J. Tang, "Mixgcf: An improved training method for graph neural network-based recommender systems," in *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*, 2021, pp. 665–674.

[4] W. Shi and R. Rajkumar, "Point-gnn: Graph neural network for 3d object detection in a point cloud," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 1711–1719.

[5] P. Pradhyumna, G. Shreya *et al.*, "Graph neural network (gnn) in image and video understanding using deep learning for computer vision applications," in *2021 Second International Conference on Electronics and Sustainable Communication Systems (ICESC)*. IEEE, 2021, pp. 1183–1189.

[6] K. Han, Y. Wang, J. Guo, Y. Tang, and E. Wu, "Vision gnn: An image is worth graph of nodes," in *Advances in Neural Information Processing Systems*, 2022.

[7] Y. Zhang, X. Yu, Z. Cui, S. Wu, Z. Wen, and L. Wang, "Every document owns its structure: Inductive text classification via graph neural networks," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2020, pp. 334–339.

[8] L. Wu, Y. Chen, H. Ji, and B. Liu, "Deep learning on graphs for natural language processing," in *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2021, pp. 2651–2653.

[9] Z. Hu, Y. Dong, K. Wang, K.-W. Chang, and Y. Sun, "Gpt-gnn: Generative pre-training of graph neural networks," in *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2020, pp. 1857–1867.

[10] D. Arya, D. K. Gupta, S. Rudinac, and M. Worring, "Hypersage: Generalizing inductive representation learning on hypergraphs," *arXiv preprint arXiv:2010.04558*, 2020.

[11] J. Huang and J. Yang, "Unignn: a unified framework for graph and hypergraph neural networks," in *Proceedings of the Thirtieth International Joint Conference on Artificial Intelligence, IJCAI-21*, 2021.

[12] E. Chien, C. Pan, J. Peng, and O. Milenkovic, "You are allset: A multiset function framework for hypergraph neural networks," in *International Conference on Learning Representations*, 2022.

[13] Y. Feng, H. You, Z. Zhang, R. Ji, and Y. Gao, "Hypergraph neural networks," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 33, 2019, pp. 3558–3565.

[14] N. Yadati, M. Nimishakavi, P. Yadav, V. Nitin, A. Louis, and P. Talukdar, "Hypergcn: A new method for training graph convolutional networks on hypergraphs," *Advances in neural information processing systems*, vol. 32, 2019.

[15] L. Sun, S. Ji, and J. Ye, "Hypergraph spectral learning for multi-label classification," in *Proceedings of the 14th ACM SIGKDD international conference on Knowledge discovery and data mining*, 2008, pp. 668–676.

[16] C. Yang, R. Wang, S. Yao, and T. Abdelzaher, "Semi-supervised hypergraph node classification on hypergraph line expansion," in *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*, 2022, pp. 2352–2361.

[17] Y. Yan, Y. Chen, S. Wang, H. Wu, and R. Cai, "Hypergraph joint representation learning for hypervertices and hyperedges via cross expansion," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 8, 2024, pp. 9232–9240.

[18] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *International Conference on Learning Representations*, 2017.

[19] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Lio, and Y. Bengio, "Graph attention networks," in *International Conference on Learning Representations*, 2018.

[20] K. Xu, W. Hu, J. Leskovec, and S. Jegelka, "How powerful are graph neural networks?" in *International Conference on Learning Representations*, 2018.

[21] F. Tudisco, K. Prokopchik, and A. R. Benson, "A nonlinear diffusion method for semi-supervised learning on hypergraphs," *arXiv preprint arXiv:2103.14867*, 2021.

[22] Y. Dong, W. Sawin, and Y. Bengio, "Hnhn: Hypergraph networks with hyperedge neurons," *arXiv preprint arXiv:2006.12278*, 2020.

[23] S. Bai, F. Zhang, and P. H. Torr, "Hypergraph convolution and hypergraph attention," *Pattern Recognition*, vol. 110, p. 107637, 2021.

[24] P. Wang, S. Yang, Y. Liu, Z. Wang, and P. Li, "Equivariant hypergraph diffusion neural operators," in *International Conference on Learning Representations (ICLR)*, 2023.

[25] I. Duta, G. Cassarà, F. Silvestri, and P. Liò, "Sheaf hypergraph networks," 2023.

[26] Z. Liu, B. Tang, Z. Ye, X. Dong, S. Chen, and Y. Wang, "Hypergraph transformer for semi-supervised classification," in *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2024, pp. 7515–7519.

[27] Y. Gao, Y. Feng, S. Ji, and R. Ji, "Hgnn+: General hypergraph neural networks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 3, pp. 3181–3199, 2023.

[28] J. H. Fowler, "Legislative cosponsorship networks in the us house and senate," *Social networks*, vol. 28, no. 4, pp. 454–465, 2006.

[29] P. S. Chodrow, N. Veldt, and A. R. Benson, "Hypergraph clustering: from blockmodels to modularity," *arXiv e-prints*, pp. arXiv–2101, 2021.

[30] Z. Wu, S. Song, A. Khosla, F. Yu, L. Zhang, X. Tang, and J. Xiao, "3d shapenets: A deep representation for volumetric shapes," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2015, pp. 1912–1920.

[31] D.-Y. Chen, X.-P. Tian, Y.-T. Shen, and M. Ouhyoung, "On visual similarity based 3d model retrieval," in *Computer graphics forum*, vol. 22, no. 3. Wiley Online Library, 2003, pp. 223–232.

[32] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *3rd International Conference on Learning Representations, ICLR 2015, San Diego, CA, USA, May 7-9, 2015, Conference Track Proceedings*, 2015.